LAB 3 ANSWERS REPORT

Oliver Wuttke – WUTT0019

Date of Completion: 18/08/2026

Flinders University – Tonsley

Artificial Intelligence (COMP3742)

Table of Contents

Table of Figures

Question 1

Apply K-Means clustering to the Wine dataset and visualize the clusters using a 2D scatter plot.

Bellow in Figure 1: K-Means Cluster on Wine Dataset shows the K-Means clustering scatters plots for the wine dataset with $k = 3$.

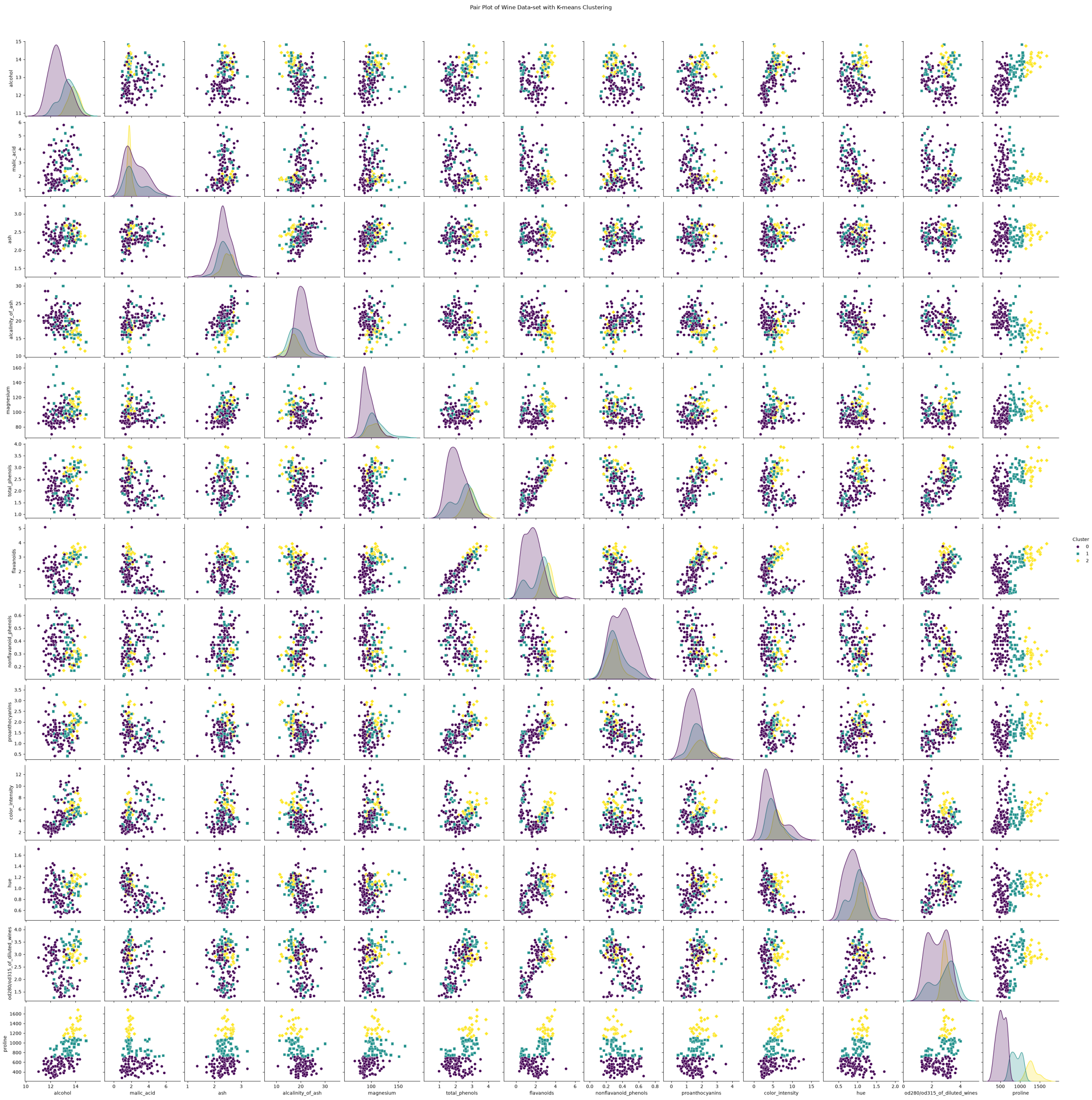

Figure 1: K-Means Cluster on Wine Dataset

Question 2

Calculate the silhouette score for the K-Means clustering performed on the Wine dataset.

The calculated silhouette score was 0.5595823478987213 for $k = 3$ clusters and no feature scaling or any hyperparameters set to non-default values.

Question 3

Apply K-Means clustering to the Wine dataset, determine the optimal number of clusters using the Elbow method, and visualize the clusters via plots. Compare the results obtained with those found using the silhouette score.

Bellow in Figure 2: Elbow Curve for K-Means in K Range [2,11] you can see that in Figure 2: Elbow Curve for K-Means in K Range [2,11] that the optimal k (highest silhouette score) is $k = 2$.

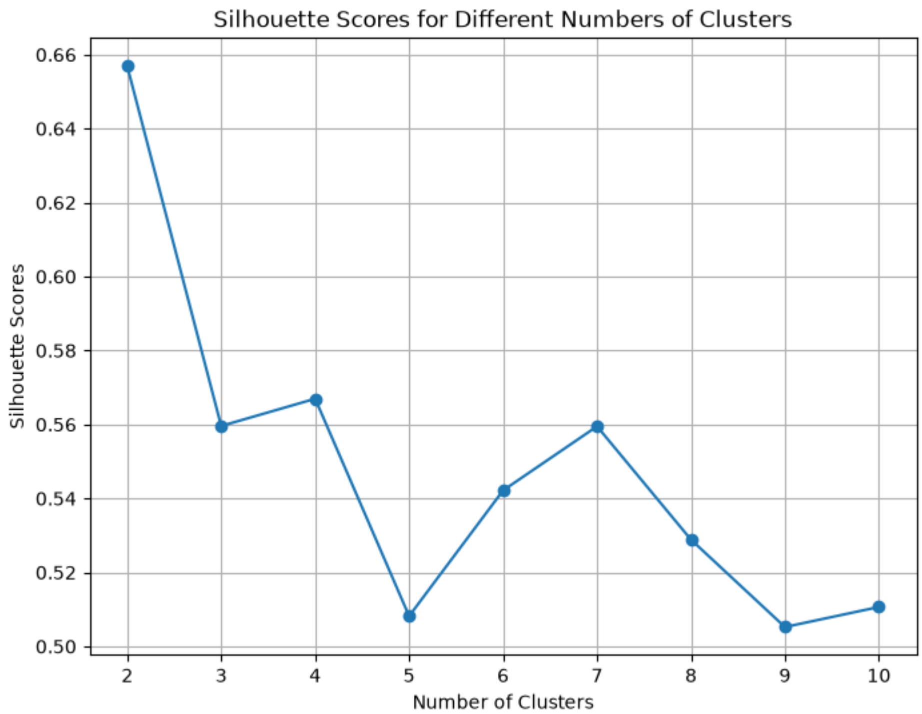

Figure 2: Elbow Curve for K-Means in K Range [2,11]

Question 4

Perform K-Means clustering on the Wine dataset, determine the optimal number of clusters using the Elbow method, and then use cluster centroids to classify new, unseen data points. Evaluate the classification accuracy and discuss any potential improvements.

With $k = 2$ for the number of clusters we were able to achieve a classification accuracy of 0.6666666666666666. This seems to be ok but we should maybe look into different hyperparameters and feature scaling to see if we can improve this accuracy.

Question 5

Apply K-Means clustering to the Wine dataset. Experiment with different feature scaling methods (e.g., StandardScaler, MinMaxScaler) and K-Means initialization techniques (e.g., k-means++, random). Compare the results using silhouette scores and visualize the clusters using pair plots.

The best silhouette score when trying different feature scaling and initialization techniques was the following configuration:

Scaler: None
Init Method: k-means++
Silhouette Score: 0.6568536504294317
Inertia: 4543749.614531862